Group Project: EDA Checklist and Rubric

Due date: Nov 20

There is a **strict** 10 page limit. This includes code, tables, text **and** plots (no title page is required; you're expected to use basic unstylized Rmd pdf output)! Your submission *must* be a pdf, and it must show all of the code (never use `echo=FALSE` in your Rmd). Failure to show the code will result in a zero in *all* relevant rubric items.

There is a strict limit on group members: 3 to 5 people. No more, no less. If you are in a group of 2 or fewer, you will be randomly added to another group of 1 or 2 people.

I strongly recommend the `patchwork` library to put plots side-by-side and thus take up less room: https://patchwork.data-imaginist.com/articles/patchwork.html. You can also control the size of plots using code chunk options as shown in the example code below.

````
```{r fig.height=6, fig.width=12}
suppressPackageStartupMessages({
    library(tidyverse)
    library(patchwork)
    theme_set(theme_bw())
})
my_df <- read_csv("path_to_data_frame")

# Plotting code
gg1 <- ggplot()
gg2 <- ggplot()

gg1 + gg2 # These plots will be side-by-side; requires the patchwork library.
my_df # DON'T DO THIS!!! THIS WILL PRINT THE DATA IN THE OUTPUT!!!
```
````

Rubric

I use a simplified rubric where I only describe levels 2 and 4; level 1 and 3 are interpreted to be "insufficient" and "between 2 and 4", respectively. Getting a level 4 in a category worth 2 marks means you get 2 marks; getting a level 3 in that category would be worth 1.5 marks, etc.

You must complete **either** Option 1 (Advanced Data Cleaning) or Option 2 (Advanced Plotting). There are no bonus marks for doing both, but you will lose marks if you do neither.

- **Choice of Data**
  - **Target Variable** [2 marks]
    - Level 2: The target variable is continuous
    - Level 4: It is abundantly clear why you chose a particular variable as the target variable.
  - **Features** [2 marks]
    - Level 2: There are one or two categorical features and one or two continuous features
    - Level 4: There are at least two categorical and continuous features that will clearly be relevant to the response.
- **Organization and Structure**
  - **Formatting and Structure** [2 marks]
    - Level 0: Rmd not properly knitted (it is completely fine if you knit to HTML or DOCX, then print to PDF).
    - Level 2: No obvious formatting issues, follows the structure in the example project.
    - Level 4: Formatting makes the project easy to read; code cells, plots, and text flow logically from one to another.
  - **Tomfoolery** [1 mark]
    - Level 0: The students made the text smaller and messed with the margins in order to fit within the 10 page limit.
    - Level 4: The output is the default PDF output from an Rmd (or is knitted to HTML and then printed to PDF) and within the 10 page limit.
- **Introduction** [4 marks]
  - Level 2: Introduction provides context for the data.
  - Level 4: Introduction makes the reader interested in the study being presented, without directly referencing the study.
- **Goals/Research Question** [4 marks]
  - Level 2: The goal of the study can be determined.
  - Level 4: The goal is written concisely and it is clear exactly what criteria are necessary to conclude that the goal is achieved.
- **Basic Data Cleaning**
  - **Checklist 1** [4 marks]

    - Level 2: At least 4 of the items in Checklist 1 are used in a way that is relevant to the exploration/modelling
    - Level 4: At least 6 of the items in Checklist 1 are used in a way that is relevant to the exploration/modelling
  - **Description** [2 marks]
    - Level 0: You simply describe each line of code
    - Level 2: The description of the cleaning is correct, but makes some references to the actual functions used.
    - Level 4: The description of the cleaning is concise, with absolutely no references to the exact function or column names. The reader could implement your cleaning process in another programming language based on your description.

- **Advanced Data Cleaning** (Option 1 of 2; you only need to do one option)
  - If you consider your cleaning to be advanced, explicitly mention it!
  - **Relevance** [2 marks]
    - Level 2: The results of the advanced cleaning are used in the exploratory plots.
    - Level 4: The results of the advanced cleaning are *necessary* for the exploratory plots.
  - **Advanced-ness** [2 marks]
    - Level 2: The cleaning is above what is in Checklist 1.
    - Level 4: The cleaning requires tidyverse functions beyond what we've done in class, or requires very careful consideration of the features in the data.

- **Advanced Plotting** (Option 2 of 2; you only need to do one option)
  - If you consider your plotting to be advanced, explicitly mention it! This applies to one of your plots; you only need to make three plots total even if you choose this option.
  - **Relevance** [2 marks]
    - Level 2: Plots are relevant to the goal.
    - Level 4: The advanced-ness of the plot contributes to the goal in a way that simpler plots would not have.
  - **Advanced-ness** [2 marks]
    - Level 2: The plotting is above what is in Checklist 2 (not just a different plot type).
    - Level 4: The plots require advanced plotting concepts, likely including a separate data frame to add context or separate data cleaning just for the plot.

- **Checklist 2** [4 marks]
  - Level 2: Across all three plots, at least 5 of the items in Checklist 2 are used in a way that is relevant to the exploration.

  - Level 4: Across all three plots, at least 8 of the items in Checklist 2 are used in a way that is relevant to the exploration.

- **Exploratory Plot 1**
  - **Content** [2 marks]
    * Level 2: The plot shows the relationship between columns
    * Level 4: The relationships are relevant to the research question.
  - **Results** [2 marks]
    * Level 2: The relationships found are discussed.
    * Level 4: The relevance to the research question is discussed clearly and concisely.
- **Exploratory Plot 2**: See Exploratory Plot 1.
- **Exploratory Plot 3**: See Exploratory Plot 1.

Checklist 1 (Data Cleaning)

☐ `mutate()`
☐ `group_by()` and `summarise()`
☐ `pivot_wider()` and/or `pivot_longer()`
☐ Some sort of regular expression in the data cleaning.
☐ Convert strings to factors, as necessary
☐ Relevel factors, such as grouping some to an "other" category or using a more reasonable reference category
☐ Convert something to date, time, or date-time.

Checklist 2 (Plotting)

☐ `geom_point()`
☐ `geom_histogram()` and/or `geom_density()` and/or `geom_boxplot()`,
☐ `geom_violin`, and/or `geom_density_ridges`
☐ `geom_bar()` and/or `geom_col()`
☐ `geom_label()` or `geom_label_repel()`
☐ `aes(colour = column_name)` or `aes(fill = column_name)`
☐ `facet_wrap()` and/or `facet_grid()`
☐ Meaningful subtitle
☐ Figure captions (either using Rmd or in the "caption" argument to `labs()` in `ggplot2`)